

Diet and asthma, allergic rhinoconjunctivitis and atopic eczema symptom prevalence: an ecological analysis of the International Study of Asthma and...

Several studies have suggested that the increasing prevalence of symptoms of asthma, rhinitis and eczema, could be associated with dietary factors. In the present paper, a global analysis of prevalence rates of wheeze, allergic rhinoconjunctivitis and atopic eczema was performed in relation to diet, as defined by national food intake data. Analyses were based on the International Study of Asthma and Allergies in Childhood (ISAAC) data for 6-7 and 13-14 yr old children. Symptoms of wheeze, allergic rhinoconjunctivitis and atopic eczema symptom prevalence were regressed against per capita food intake, and adjusted for gross national product to account for economic development. Dietary data were based on 1995 Food and Agriculture Organisation of the United Nations data for 53 of the 56 countries that took part in ISAAC phase I (1994/1995). The 13-14 year age group showed a consistent pattern of decreases in symptoms of wheeze (current and severe), allergic rhinoconjunctivitis and atopic eczema, associated with increased per capita consumption of calories from cereal and rice, protein from cereals and nuts, starch, as well as vegetables and vegetable nutrients. The video questionnaire data for 13-14 yr olds and the ISAAC data for 6-7 yr olds showed similar patterns for these foods. A consistent inverse relationship was seen between prevalence rates of the three conditions and the intake of starch, cereals, and vegetables. If these findings could be generalised, and if the average daily consumption of these foods increased, it is speculated that an important decrease in symptom prevalence may be achieved.